COMPASS Synoptic: Porewater SO4/Cl

July 2026 Samples

2026-10-29

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1 Run Information

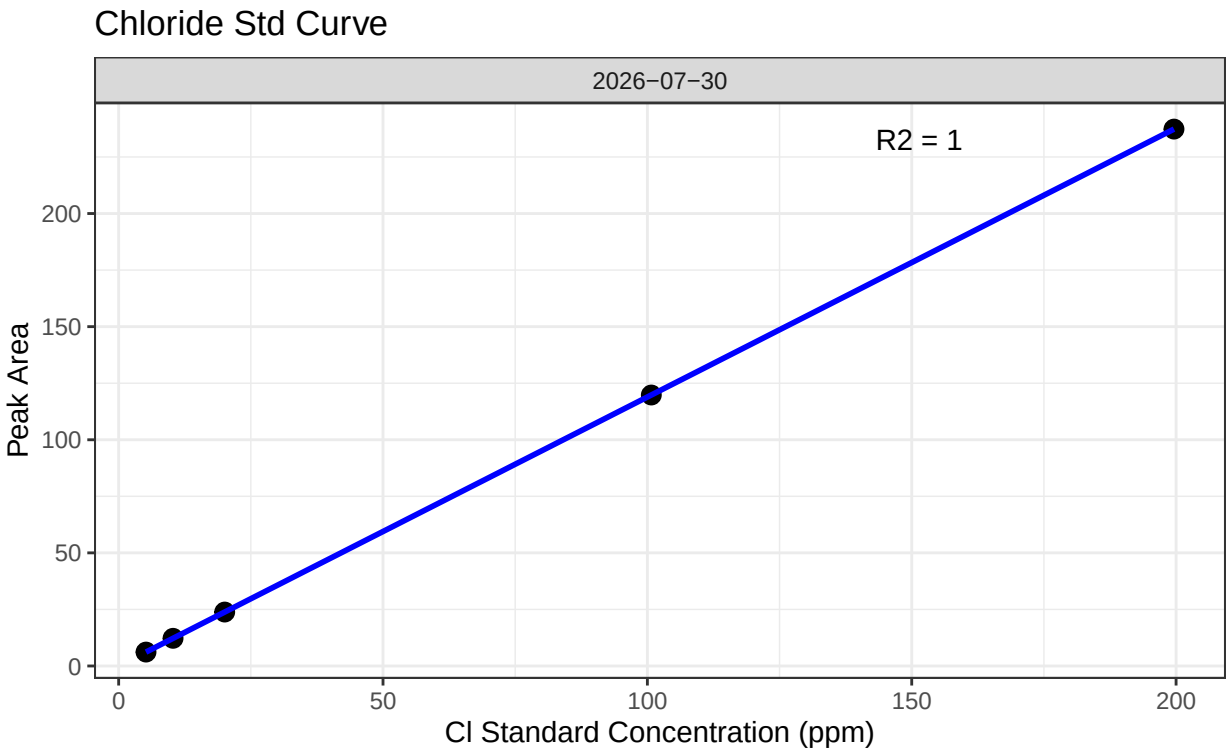
Run date: 2026-07-30
Run by: Zoe Read
Run notes: Run notes

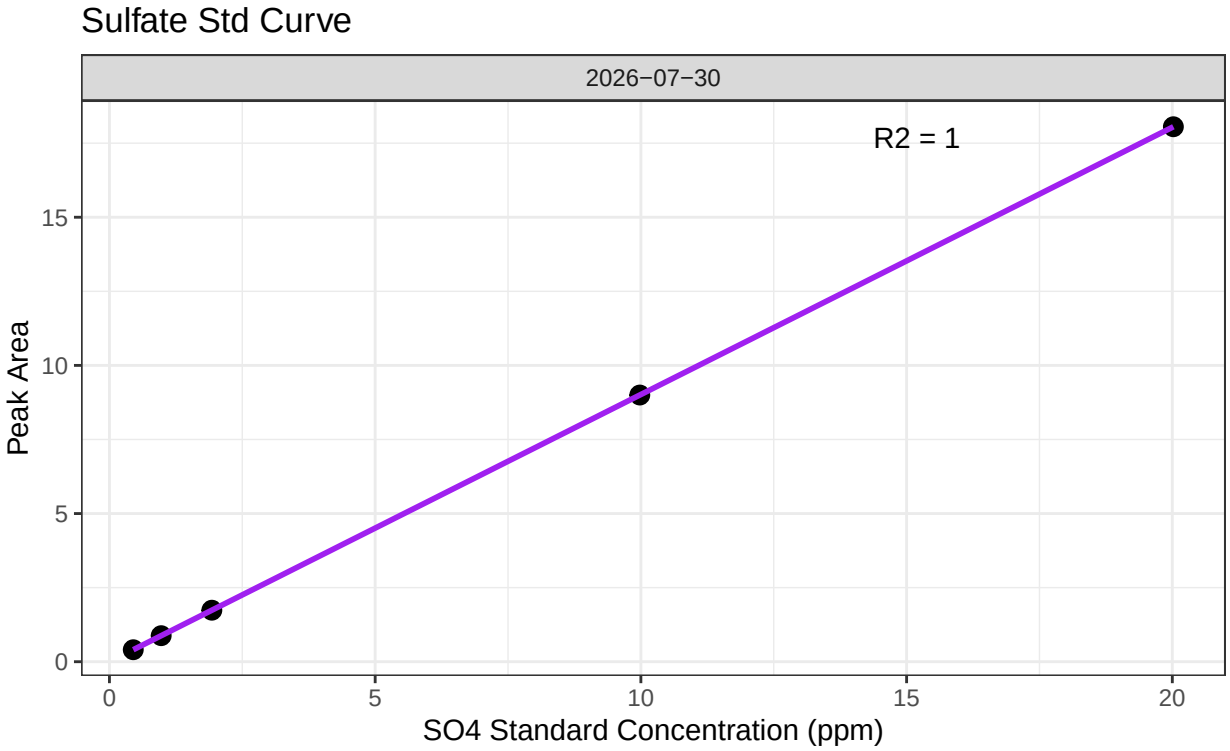
SO4 file used: - COMPASS_Synoptic_CB_MonMon_202607_MSM_GWI_SO4.txt

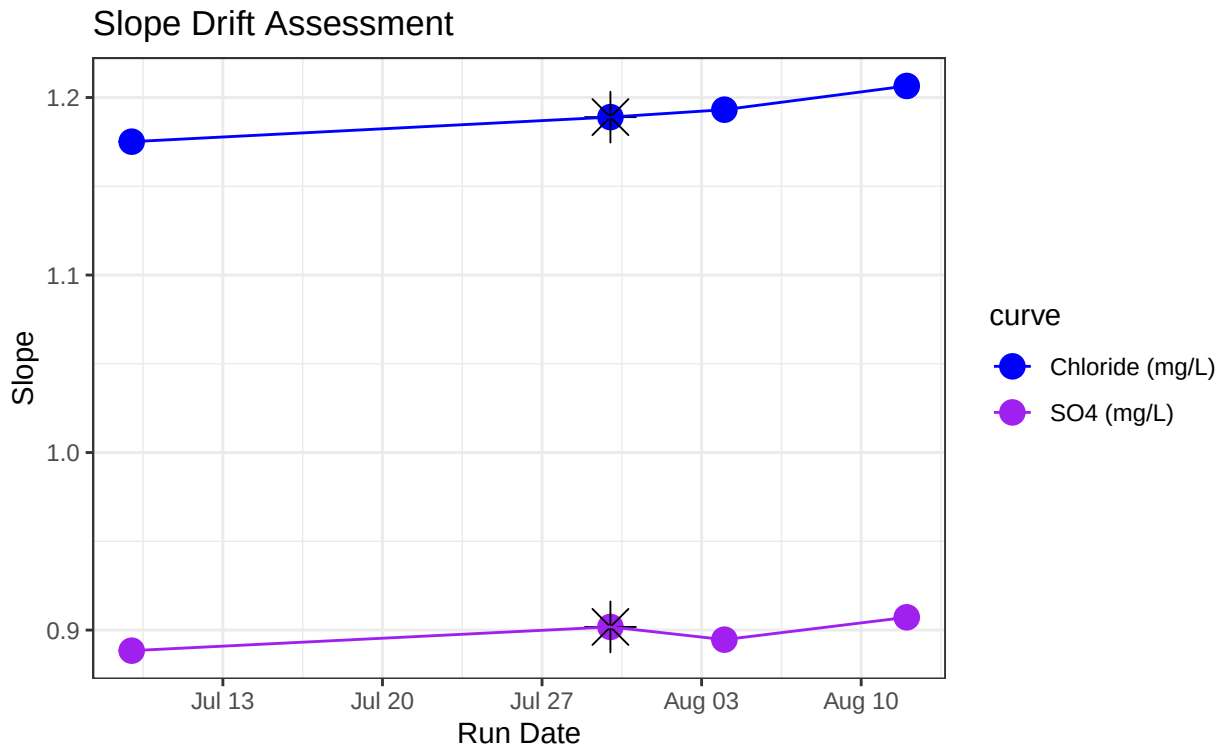
Cl file used: - COMPASS_Synoptic_CB_MonMon_202607_MSM_GWI_Cl.txt

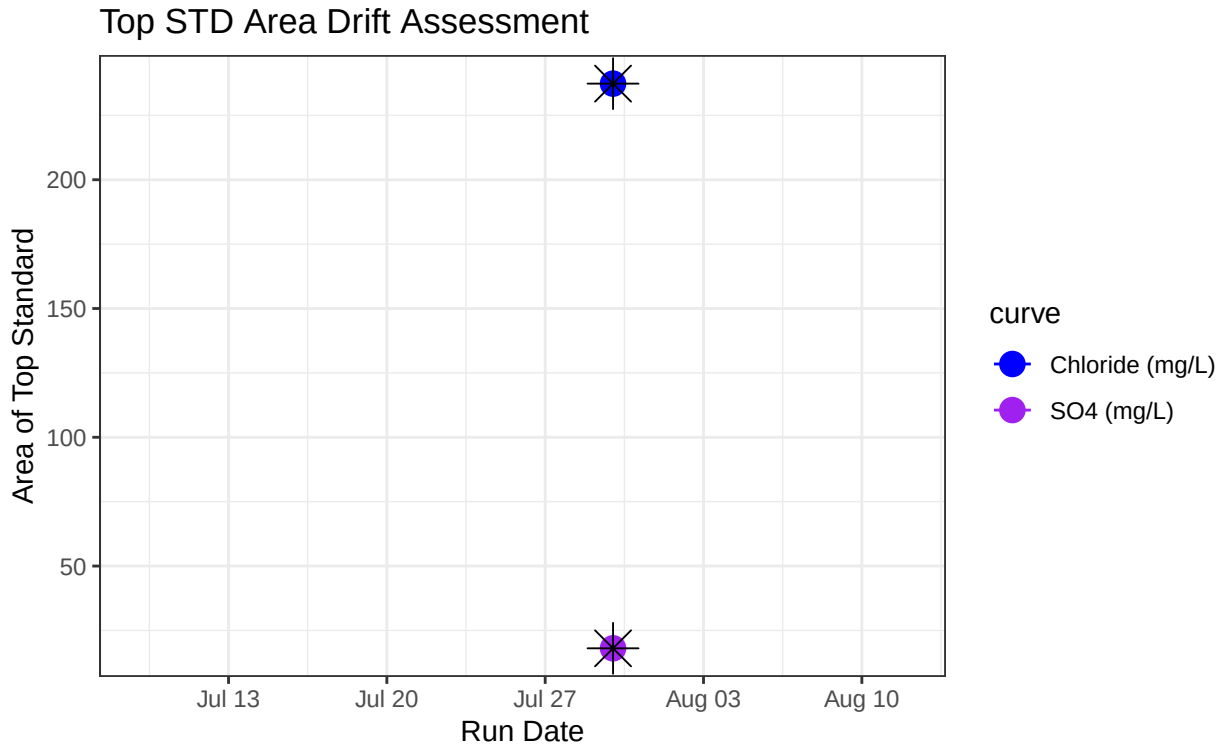
Changed sample names: `IDs_to_change` **Samples removed:** `samples_to_remove`

1.1 Assess Standard Curves










```
## [1] "PROCEED - Cl Curve r2 GOOD"
```

```
## [1] "PROCEED - SO4 Curve r2 GOOD"
```

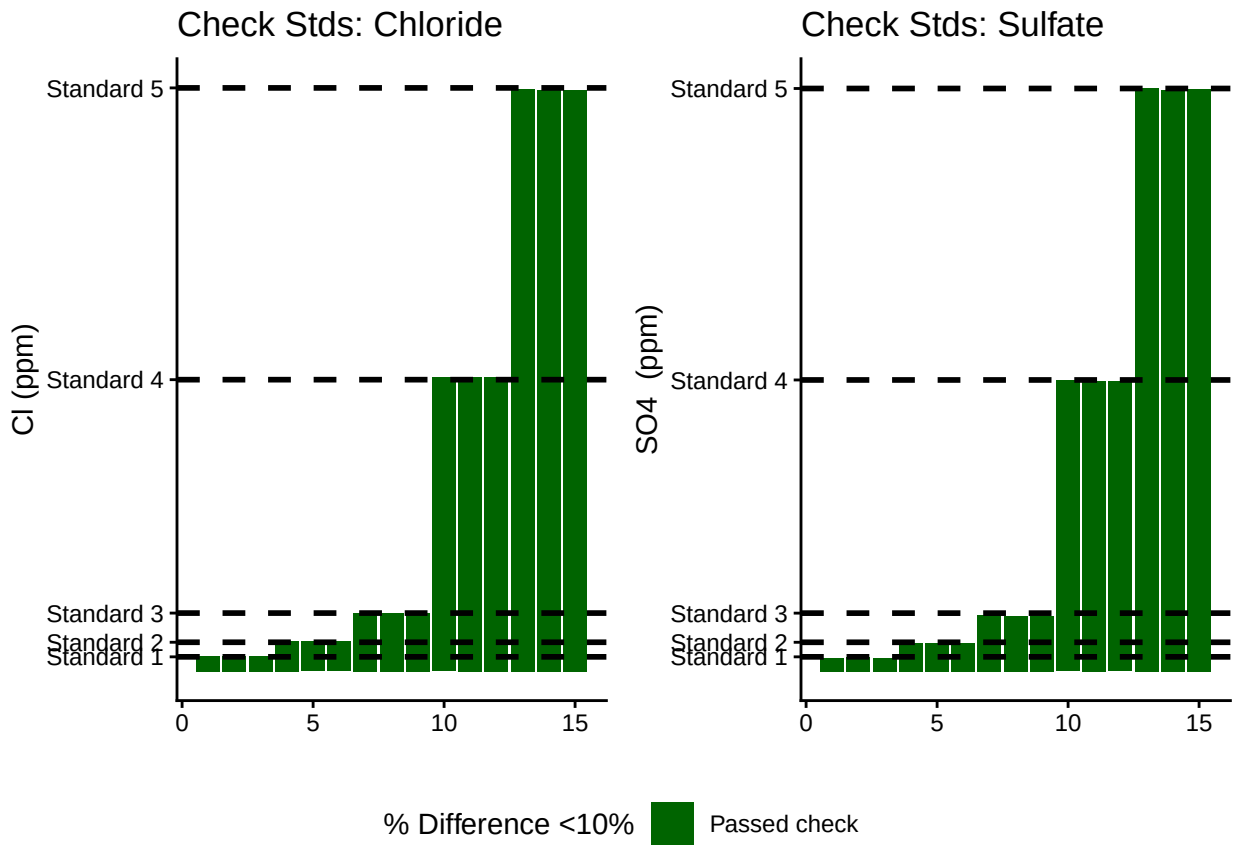
1.2 Assess Check Standards

sample_ID	mean_Cl	cv_Cl	flag_Cl
Standard 1	5.192	0.004	PROCEED - Chloride Check Standard RSD within Range
Standard 2	10.272	0.002	PROCEED - Chloride Check Standard RSD within Range
Standard 3	20.067	0.000	PROCEED - Chloride Check Standard RSD within Range
Standard 4	100.810	0.001	PROCEED - Chloride Check Standard RSD within Range
Standard 5	199.389	0.001	PROCEED - Chloride Check Standard RSD within Range

sample_ID	mean_SO4	cv_SO4	flag_SO4
Standard 1	0.459	0.042	PROCEED - Sulfate Check Standard RSD within Range
Standard 2	0.972	0.007	PROCEED - Sulfate Check Standard RSD within Range
Standard 3	1.914	0.006	PROCEED - Sulfate Check Standard RSD within Range
Standard 4	9.969	0.002	PROCEED - Sulfate Check Standard RSD within Range
Standard 5	19.985	0.002	PROCEED - Sulfate Check Standard RSD within Range

```
## [1] "PROCEED - >80% of Chloride Check Standards have RSD within range"
```

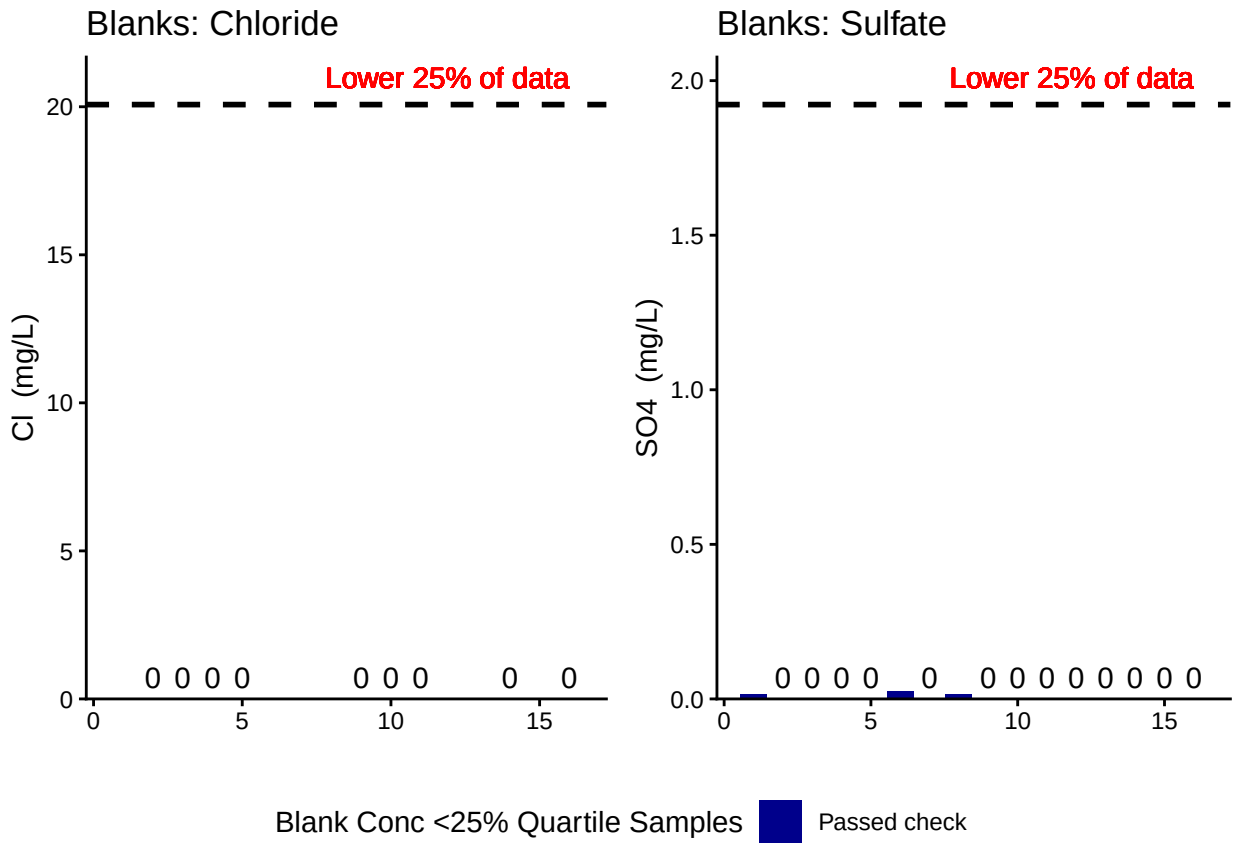
```
## [1] "PROCEED - >80% of Sulfate Check Standards have RSD within range"
```




```
## [1] "PROCEED - >80% of Chloride Check Standards are within range of expected concentration"
```

```
## [1] "PROCEED - >80% of Sulfate Check Standards are within range of expected concentration"
```

1.3 Assess Blanks

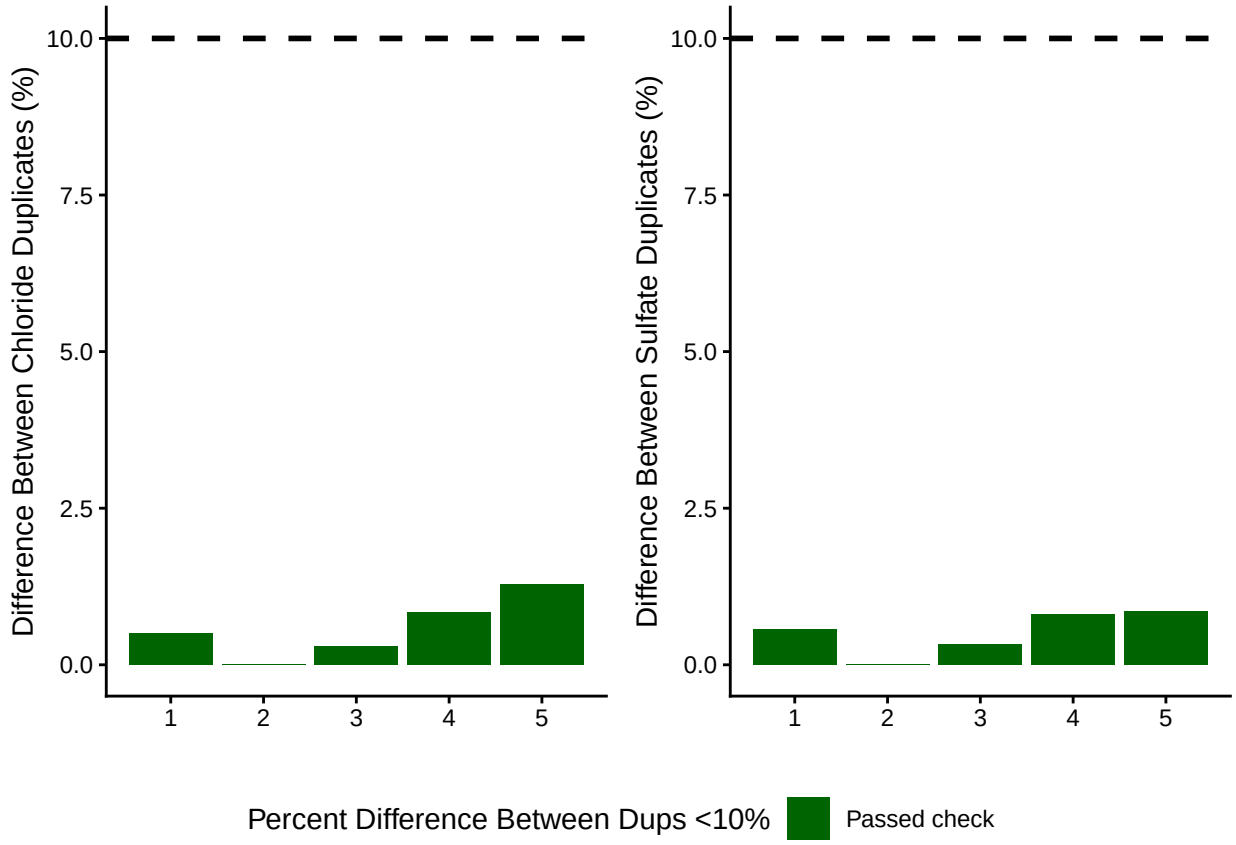


analyte	run_date	mean_blanks
Cl	2026-07-30	0.008
SO4	2026-07-30	0.003

```
## [1] "PROCEED - >80% of Chloride Blank concentrations are in the lower 25% quartile of samples"
```

```
## [1] "PROCEED - >80% of Sulfate Blank concentrations are in the lower 25% quartile of samples"
```

1.4 Assess Duplicates

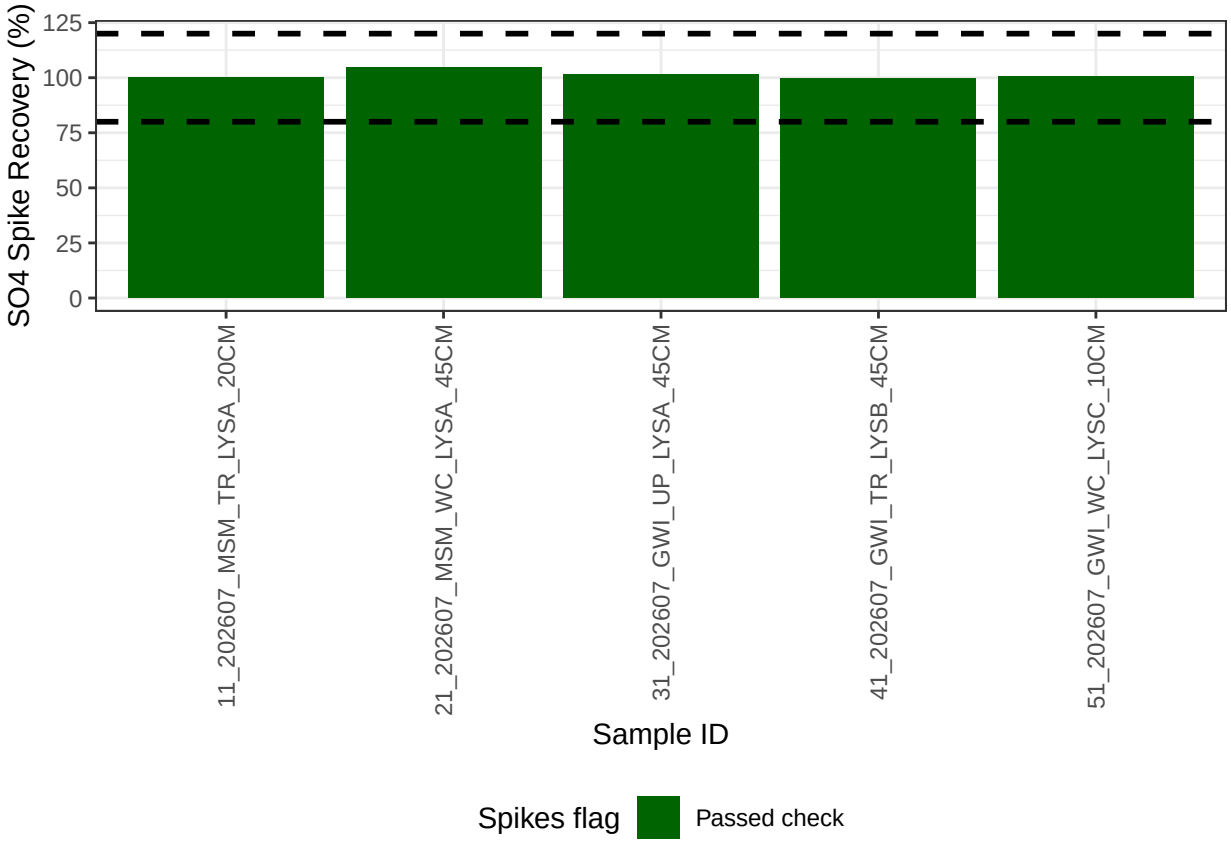



```
## [1] "PROCEED - >80% of Chloride Duplicates have a percent difference <10%"
```

```
## [1] "PROCEED - >80% of Sulfate Duplicates have a percent difference <10%"
```

1.5 Calculate mmol/L concentrations & salinity, add dilutions

1.6 Assess Analytical Spikes




```
## [1] "PROCEED - >80% of SO4 spikes have a recovery between the high and low cutoff"
```

1.7 Check if samples are within the range of the standard curve

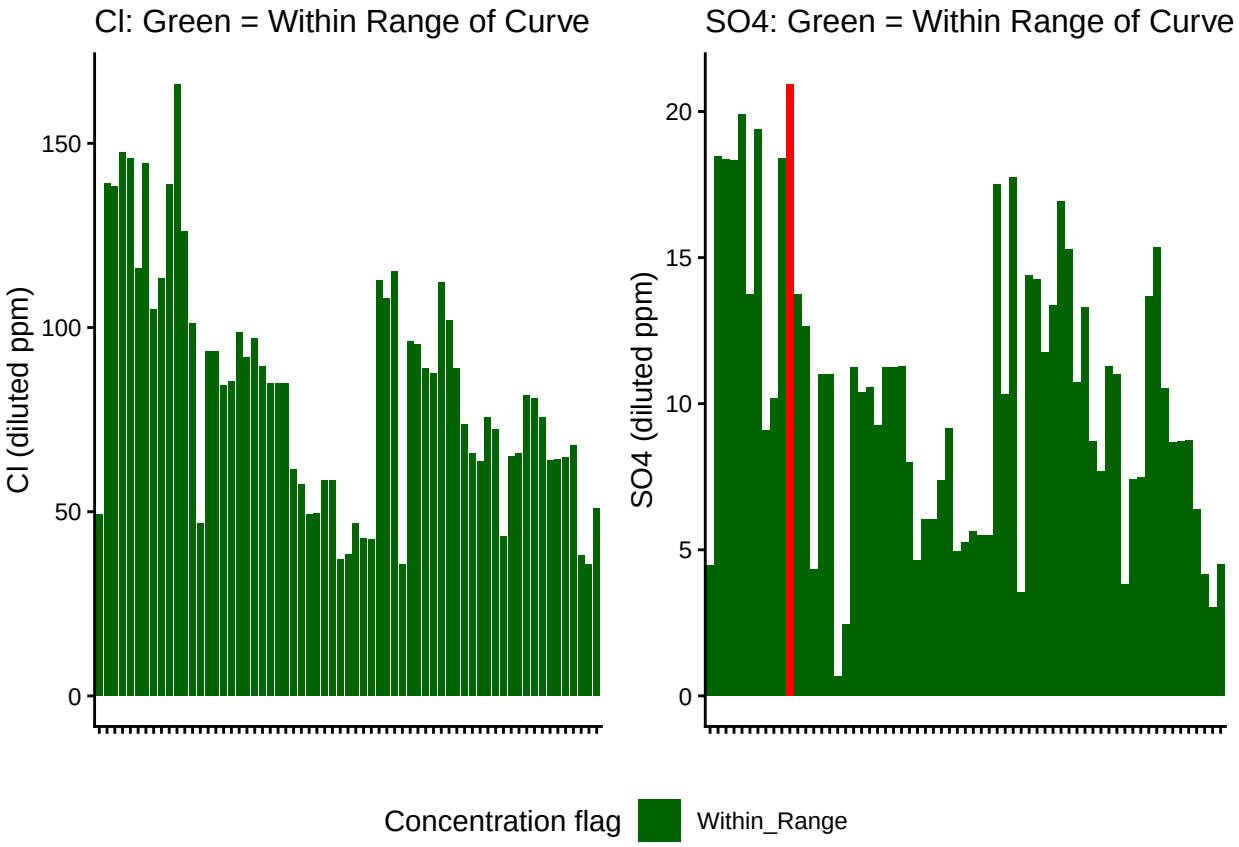


Table 4: SO4 samples

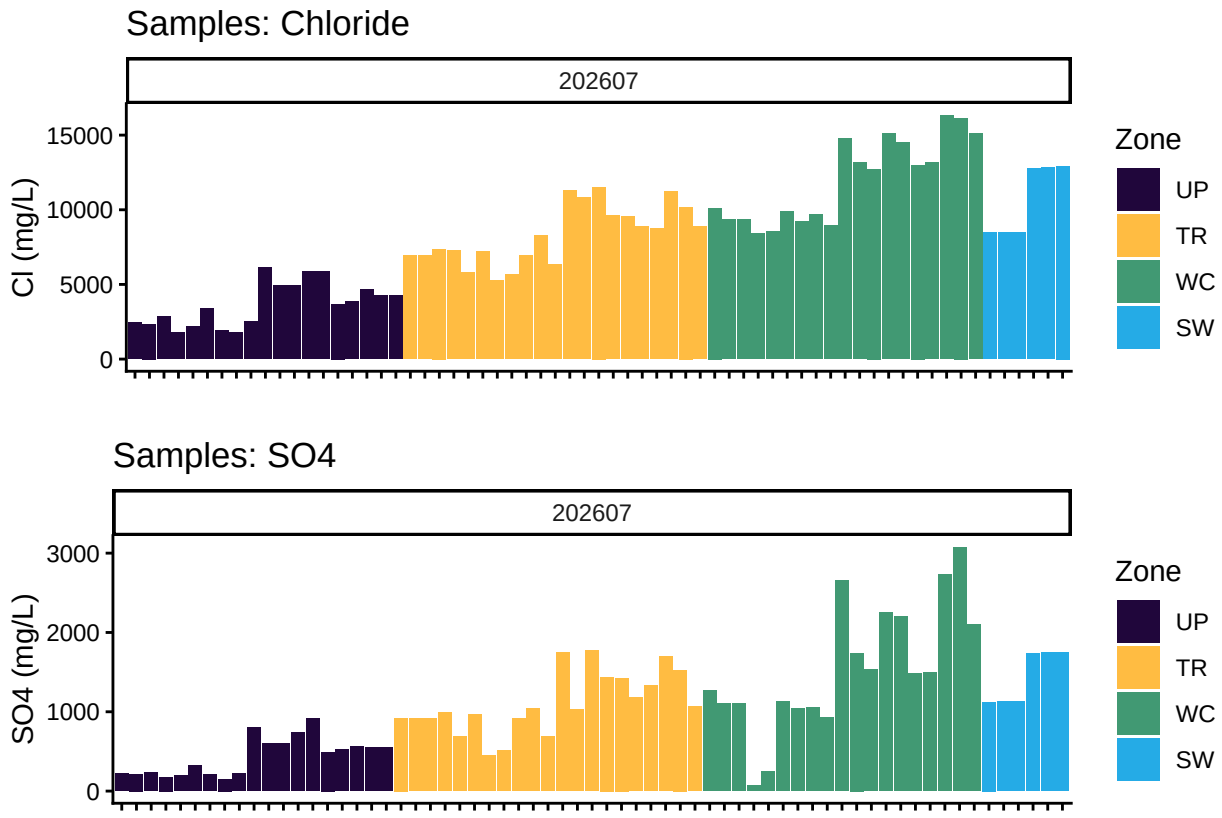
SO4_Conc_flag	Percent_samples
Within_Range	98.461539
adl	1.538461

Table 5: Cl samples

Cl_Conc_flag	Percent_samples
Within_Range	100

1.8 Check to see if samples run match metadata & merge info

```
## All sample IDs are present in metadata.
```


Samples: Chloride
202607
Cl (mg/L)
15000
10000
5000
0
Zone
UP
TR
WC
SW
Samples: SO4
202607
SO4 (mg/L)
3000
2000
1000
0
Zone
UP
TR
WC
SW

#end